

Explainable AI for Chronic Kidney Disease Prediction using Machine Learning and SHAP

Megha Kalachira Ankush

Machine Learning and Explainable AI Project

1. Introduction

Chronic Kidney Disease (CKD) is a progressive medical condition that affects kidney function over time and can lead to kidney failure if not diagnosed early. Early detection of CKD is important because timely medical intervention can reduce complications and improve patient outcomes.

Artificial Intelligence and Machine Learning techniques are increasingly being used in healthcare for disease prediction and clinical decision support. However, many machine learning models act as “black boxes,” making it difficult for clinicians to understand why predictions are made. In healthcare environments, interpretability and transparency are essential for trust and reliability.

This project presents an Explainable AI framework for Chronic Kidney Disease prediction using a Decision Tree classifier combined with SHAP (SHapley Additive Explanations). The goal is not only to achieve accurate predictions but also to provide understandable explanations for clinical decision-making.

2. Dataset Description

The Chronic Kidney Disease dataset contains clinical and laboratory measurements associated with kidney function.

Dataset Information

- Total Samples: 400
- Total Features: 26
- Target Variable: Classification (CKD / Not CKD)

Important Features

- Age
- Blood Pressure (bp)
- Specific Gravity (sg)
- Albumin (al)
- Blood Glucose Random (bgr)
- Blood Urea (bu)
- Serum Creatinine (sc)
- Sodium (sod)
- Potassium (pot)
- Hemoglobin (hemo)
- Packed Cell Volume (pcv)
- Hypertension (htn)
- Diabetes Mellitus (dm)

The target variable indicates whether the patient has Chronic Kidney Disease.

3. Data Preprocessing

Several preprocessing operations were performed before training the machine learning model.

3.1 Column Cleaning

The unnecessary id column was removed because it does not contribute to prediction.

3.2 Standardization

Column names were standardized by converting them into lowercase and removing extra spaces.

3.3 Handling Categorical Variables

Categorical values such as:

- yes/no
- present/notpresent
- normal/abnormal

were converted into numerical values for machine learning compatibility.

3.4 Missing Value Handling

The dataset contained missing values in several columns.

Examples:

- Red Blood Cells (rbc): 152 missing values
- Sodium (sod): 87 missing values
- Potassium (pot): 88 missing values

Missing values were handled using mean imputation through the SimpleImputer method.

4. Machine Learning Methodology

4.1 Feature and Target Separation

The dataset was divided into:

- Input Features (X)
- Target Variable (y)

4.2 Train-Test Split

The dataset was split into:

- 80% Training Data
- 20% Testing Data

A fixed random state was used to ensure reproducibility.

5. Decision Tree Model

A Decision Tree Classifier was selected for this project due to:

- High interpretability
- Simplicity
- Suitability for clinical decision-support systems

The model was trained using the processed training dataset.

6. Explainable AI using SHAP

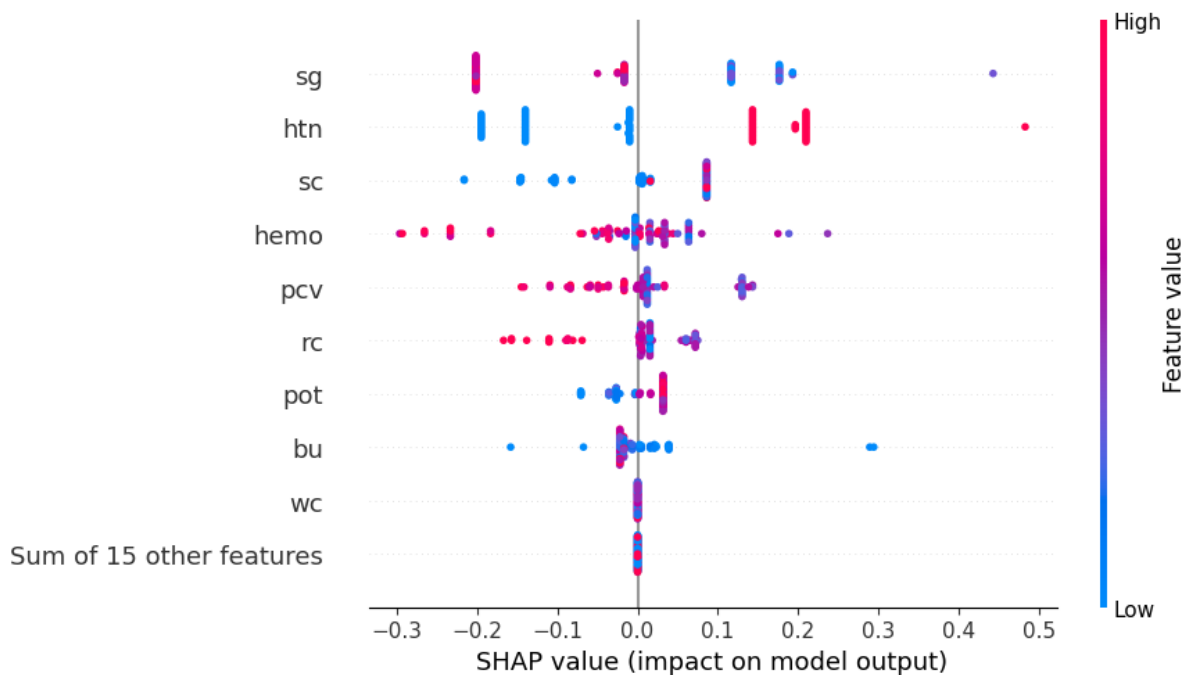
SHAP (SHapley Additive Explanations) was applied to interpret the Decision Tree predictions.

SHAP helps explain:

- Which features influenced predictions
- Whether features increased or decreased CKD risk
- Global and local model behavior

7. SHAP Visualizations

7.1 SHAP Beeswarm Plot



SHAP Beeswarm Plot

The SHAP Beeswarm plot provides a global explanation of feature importance across all predictions.

Interpretation

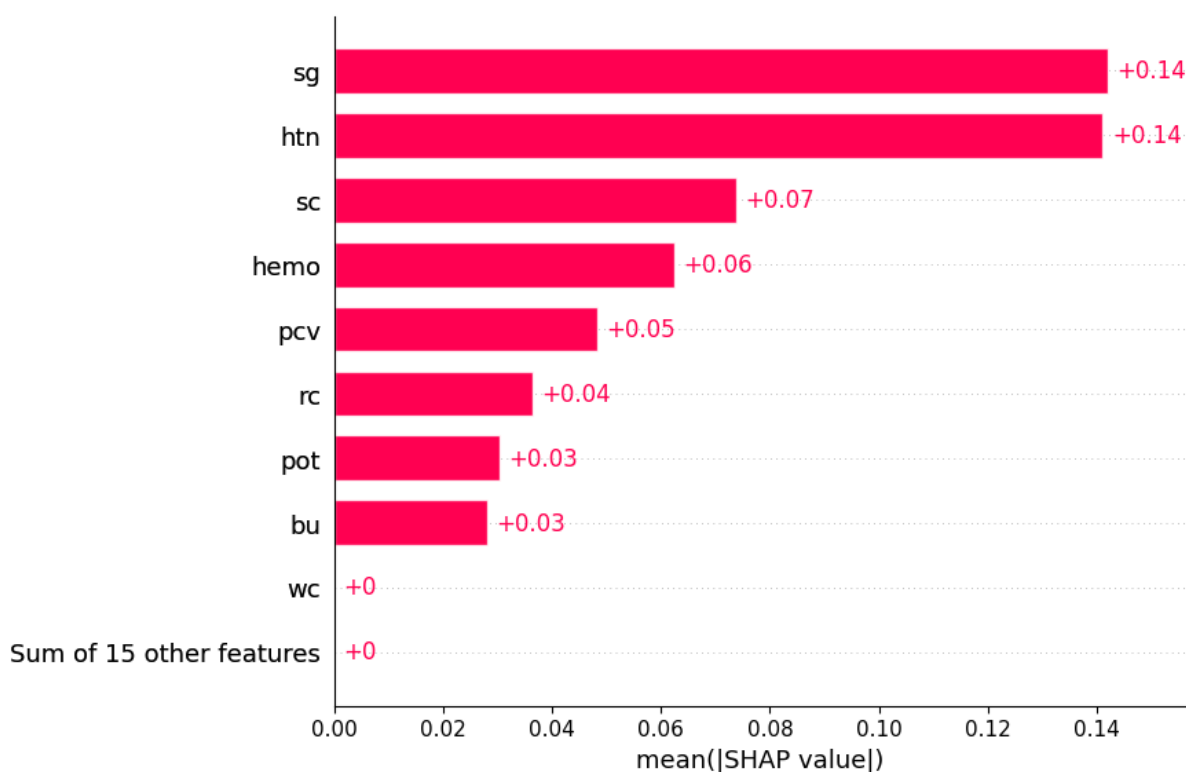
- Features at the top have higher influence on predictions.
- Red points indicate higher feature values.

- Blue points indicate lower feature values.
- Positive SHAP values push predictions toward CKD.
- Negative SHAP values push predictions away from CKD.

Key Observation

Haemoglobin (hemo) showed the strongest influence on CKD prediction.

7.2 SHAP Bar Plot



SHAP Bar Plot

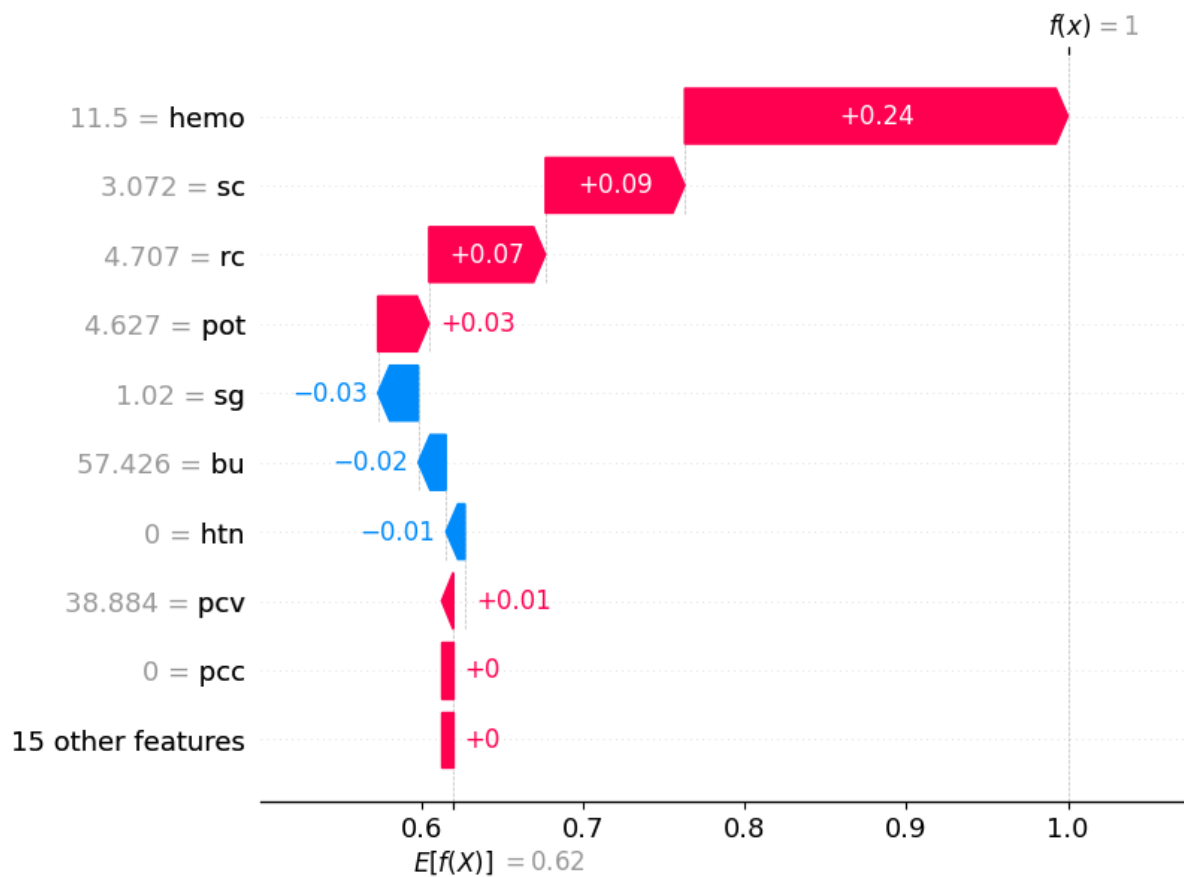
The SHAP Bar Plot summarizes average feature impact.

Most Important Features

1. Hemoglobin (hemo)
2. Specific Gravity (sg)
3. Hypertension (htn)
4. Blood Urea (bu)
5. Packed Cell Volume (pcv)

These features consistently influenced the model's predictions.

7.3 SHAP Waterfall Plot



SHAP Waterfall Plot

The Waterfall Plot explains a single patient prediction.

Interpretation

- Red arrows increase CKD probability
- Blue arrows decrease CKD probability
- The model prediction is built step-by-step from feature contributions

This visualization improves transparency in individual clinical predictions.

8. Model Evaluation

8.1 Accuracy

Training Accuracy

1.00

Testing Accuracy

1.00

The model achieved perfect classification performance on the testing dataset.

8.2 Classification Report

Classification Report:

	precision	recall	f1-score	support
0.0	1.00	1.00	1.00	28
1.0	1.00	1.00	1.00	52
accuracy			1.00	80
macro avg	1.00	1.00	1.00	80
weighted avg	1.00	1.00	1.00	80

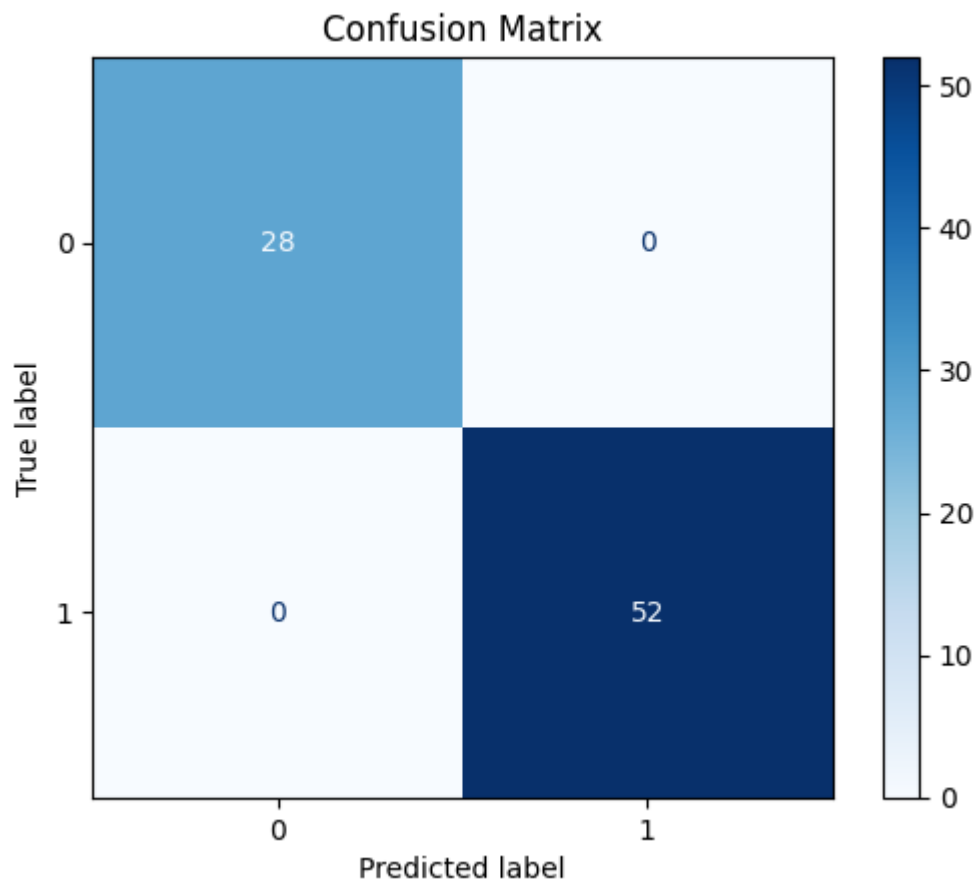
Classification Report Screenshot

Results

Metric	Score
Precision	1.00
Recall	1.00
F1-Score	1.00

The model successfully identified CKD and non-CKD cases with very high performance.

9. Confusion Matrix



Confusion Matrix

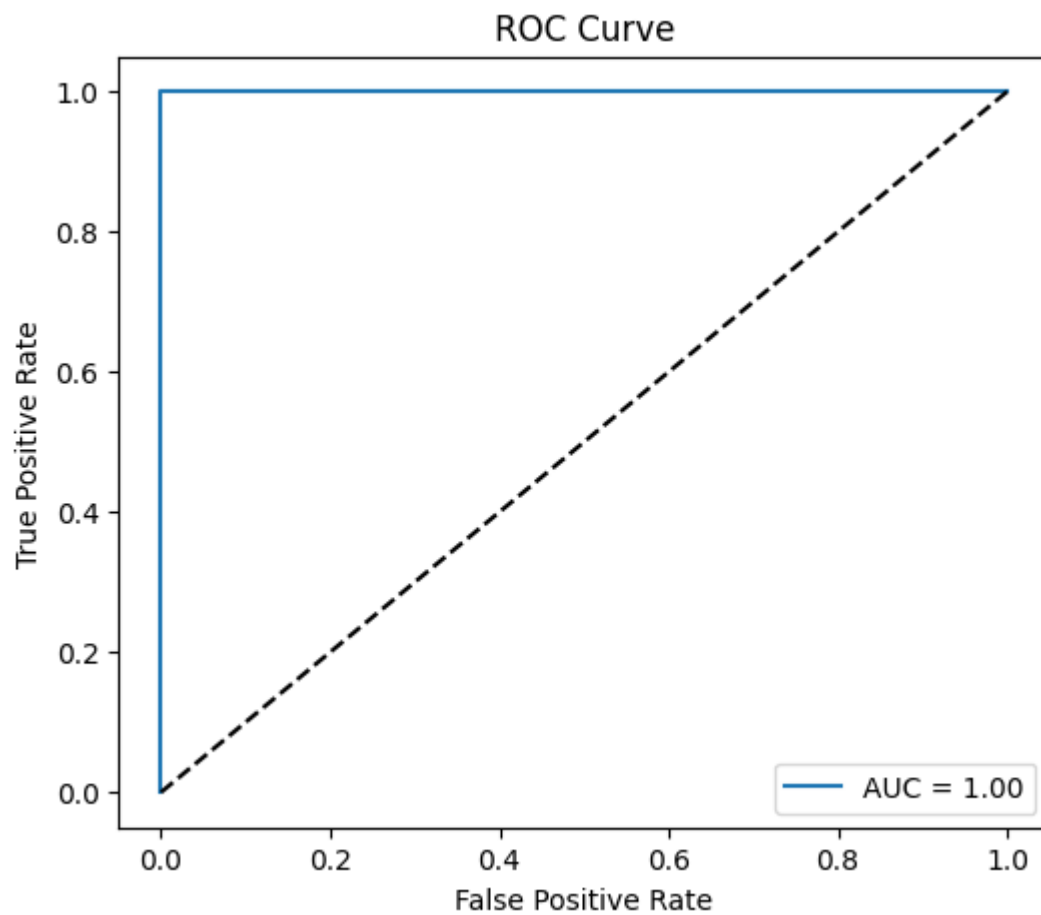
The confusion matrix shows:

- True Positives
- True Negatives
- False Positives
- False Negatives

Observation

The model produced nearly perfect predictions with no major classification errors.

10. ROC Curve and AUC Score



ROC Curve

ROC-AUC Score

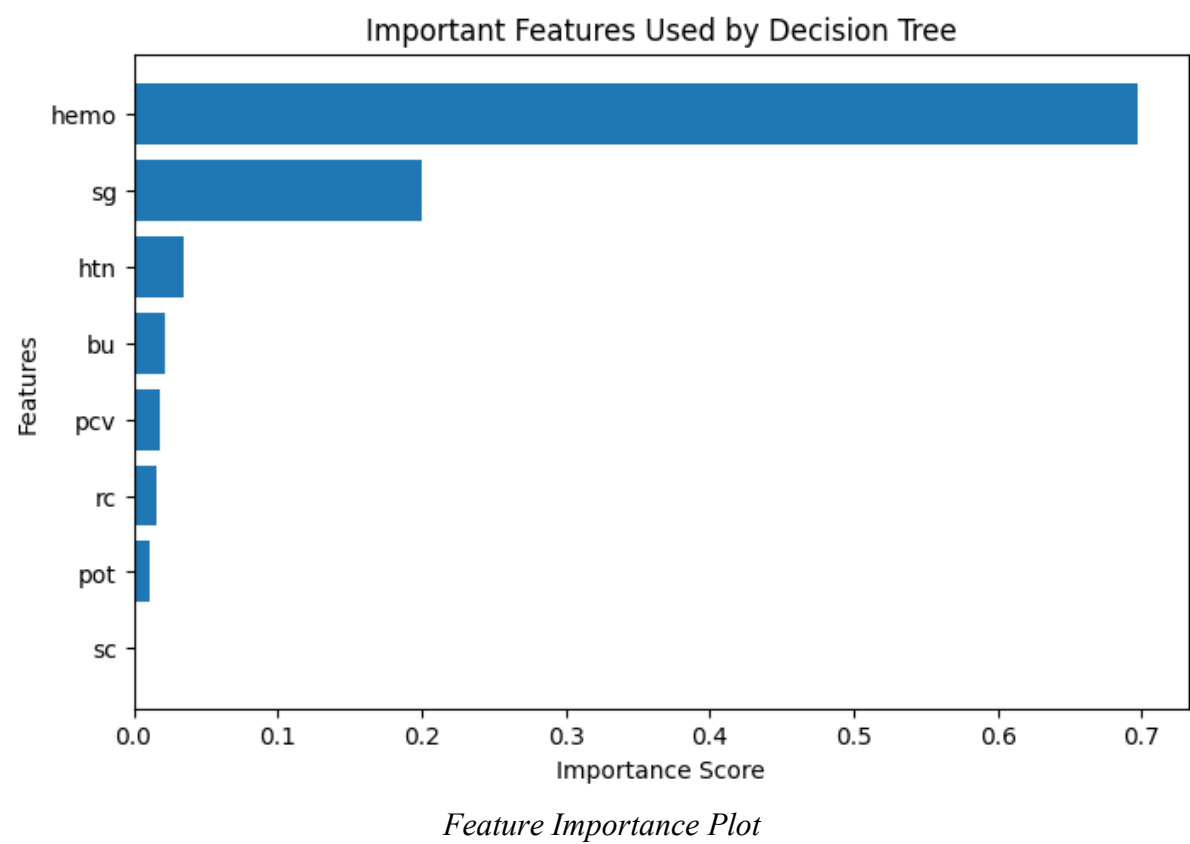
1.00

Interpretation

The ROC curve demonstrates the model's ability to distinguish between CKD and non-CKD patients.

An AUC score of 1.00 indicates excellent classification performance.

11. Feature Importance Analysis



The Decision Tree feature importance analysis identified the most influential clinical features.

Feature Importance Results

Feature	Importance
Hemoglobin (hemo)	0.697
Specific Gravity (sg)	0.199
Hypertension (htn)	0.034
Blood Urea (bu)	0.021
Packed Cell Volume (pcv)	0.018

Observation

Haemoglobin was identified as the most important predictor for CKD classification.

12. Cross Validation Analysis

Cross-validation was performed to evaluate model stability.

Cross Validation Scores

- 0.975
- 0.9875
- 0.9500
- 0.9625
- 0.9750

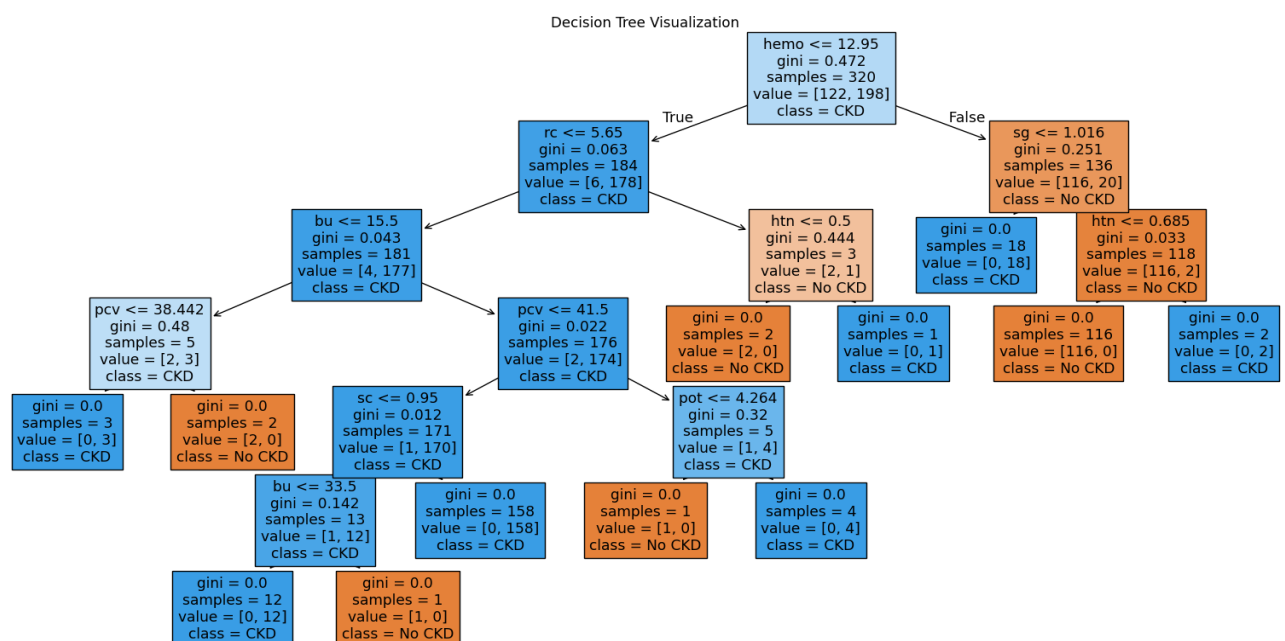
Average Cross Validation Score

0.97

Interpretation

The model demonstrated strong consistency across multiple validation folds.

13. Decision Tree Visualization



Decision Tree Visualization

The Decision Tree visualization illustrates how the model makes predictions using clinical decision rules.

The tree structure improves interpretability and supports explainable clinical reasoning.

14. Key Findings

- Hemoglobin is the strongest predictor of Chronic Kidney Disease.
- SHAP explainability improves transparency of predictions.
- The Decision Tree model achieved excellent classification performance.
- Explainable AI methods can support trustworthy healthcare systems.

15. Limitations

- The dataset size is relatively small.
- Missing values required imputation.
- Perfect accuracy may indicate possible overfitting.
- External validation using real clinical datasets is required.

16. Conclusion

This project demonstrated an Explainable AI framework for Chronic Kidney Disease prediction using a Decision Tree classifier and SHAP analysis.

The model achieved high predictive performance while maintaining transparency and interpretability. SHAP explainability techniques helped identify clinically meaningful predictors and improved understanding of machine learning decisions.

This approach can contribute toward trustworthy AI-assisted healthcare decision-support systems.

17. Future Work

Future improvements may include:

- Testing on larger hospital datasets
- Comparing multiple machine learning models
- Developing deep learning-based explainable systems
- Deploying the model as a clinical web application
- Evaluating real-world healthcare integration